

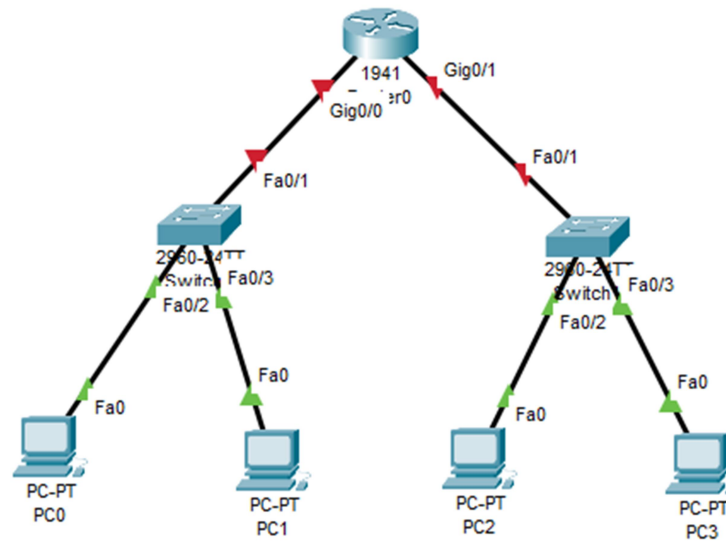


Fig: Network before assigning ip addresses

```

Router>enable
Router#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Deep
Deep(config)#interface g
Deep(config)#interface gigabitEthernet 0/0
Deep(config-if)#ip address 192.168.10.4 255.255.255.128
Deep(config-if)#no shutdown

Deep(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Deep(config-if)#exit
Deep(config)#exit
Deep#
%SYS-5-CONFIG_I: Configured from console by console

Deep#write
Building configuration...
[OK]

Deep(config)#interface GigabitEthernet0/1
Deep(config-if)#ip address 192.168.10.132 255.255.255.128
Deep(config-if)#no shutdown

Deep(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Deep(config-if)#exit
Deep(config)#exit
Deep#
%SYS-5-CONFIG_I: Configured from console by console

Deep#write
Building configuration...
[OK]

```

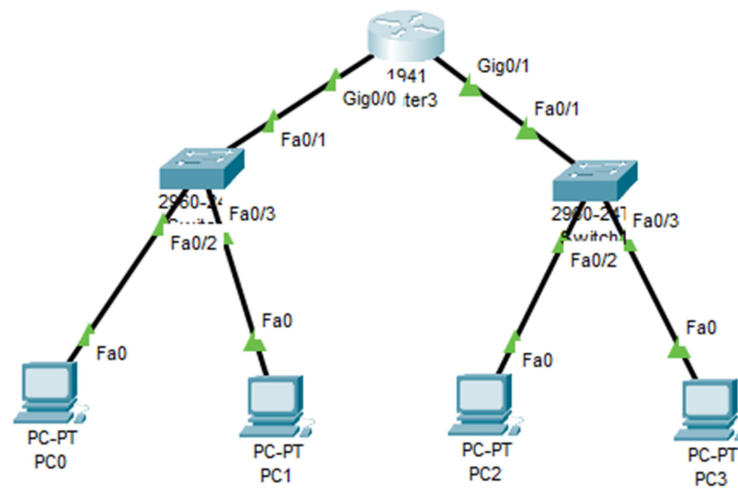


Fig: Network after assigning ip addresses

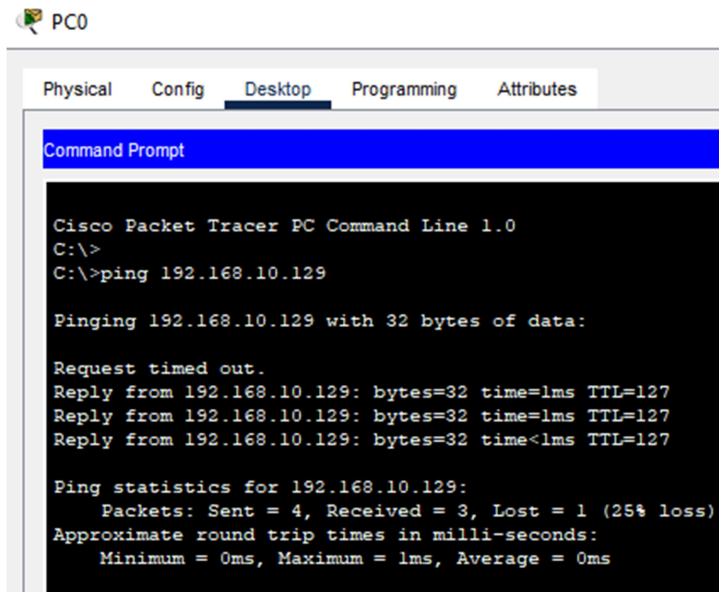
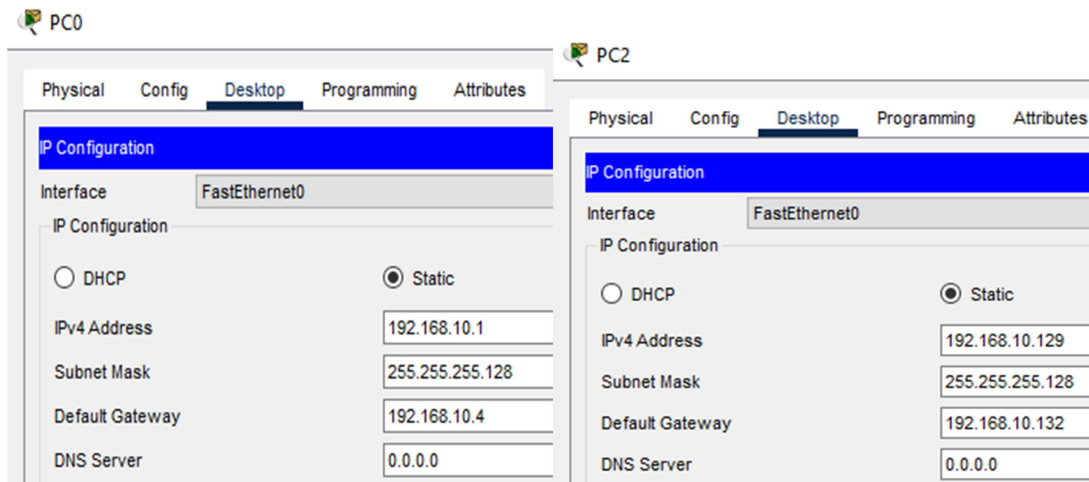


Fig: ping test between PC0 and PC2

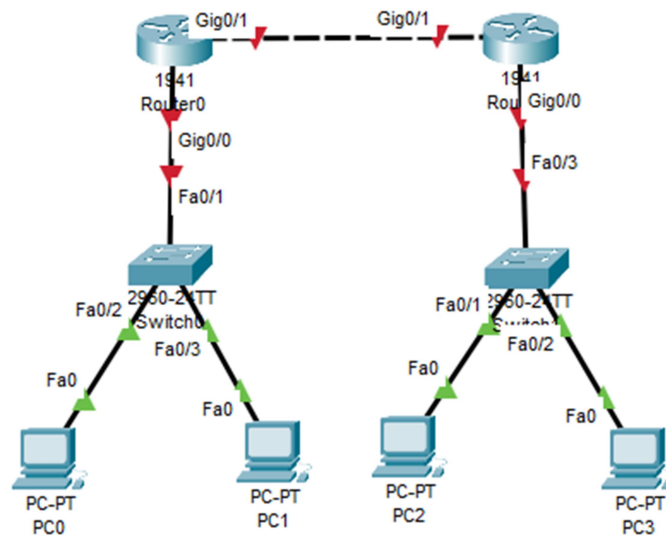


Fig: Network topology

PC0

Physical	Config	Desktop	Programming	Attributes
IP Configuration				
Interface: FastEthernet0				
IP Configuration				
☐ DHCP ☒ Static				
IPv4 Address: 192.168.2.7				
Subnet Mask: 255.255.255.0				
Default Gateway: 192.168.2.3				
DNS Server: 0.0.0.0				

```

Deep#en
Deep#conf t
Deep#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Deep(config)#ip route 192.168.3.0 255.255.255.0 192.168.1.4
Deep(config)#ex
Deep#
%SYS-5-CONFIG_I: Configured from console by console

Deep#write
Building configuration...
[OK]
          
```

PC2

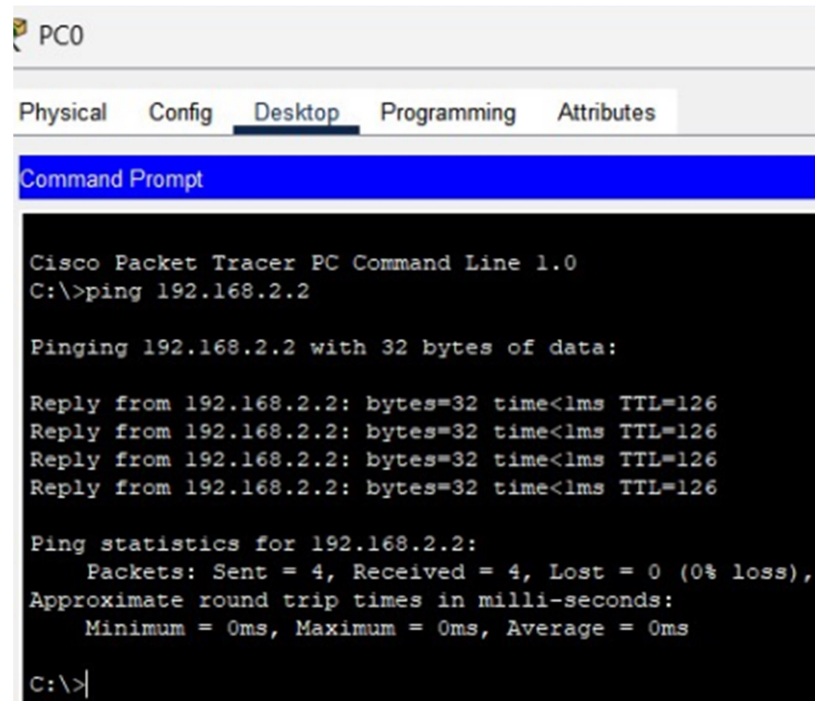
Physical	Config	Desktop	Programming	Attributes
IP Configuration				
Interface: FastEthernet0				
IP Configuration				
☐ DHCP ☒ Static				
IPv4 Address: 192.168.3.5				
Subnet Mask: 255.255.255.0				
Default Gateway: 192.168.3.3				
DNS Server: 0.0.0.0				

Fig: ip routing in router0

```

Deepl#en
Deepl#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Deepl(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.2
Deepl(config)#ex
Deepl#
%SYS-5-CONFIG_I: Configured from console by console
          
```

Fig:ip routing in router1



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of a ping command to the IP address 192.168.2.2. The output indicates that four packets were sent and received successfully, with a 0% loss rate and a round trip time of 0ms.

```
PC0
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

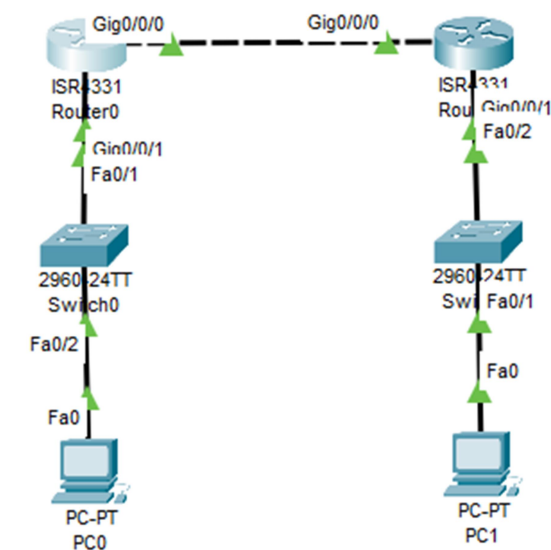
Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig: Ping test between PC1 and PC3

Conclusion

This lab demonstrated how static routing enables communication between different networks by manually configuring routes on each router. While effective for small networks, it also showed that static routing is not scalable and requires manual updates when the network changes.



Router0

Physical Config CLI Attributes

IOS Command Line Interface

```

Deep(config-if)#router rip
Deep(config-router)#network 192.168.1.0
Deep(config-router)#network 192.168.3.0
Deep(config-router)#end
Deep#
%SYS-5-CONFIG_I: Configured from console by console
          
```

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```

Deep(config-if)#
Deep(config-if)#exit
Deep(config)#router rip
Deep(config-router)#no network 192.168.3.0
Deep(config-router)#network 192.168.3.0
Deep(config-router)#end
Deep#
%SYS-5-CONFIG_I: Configured from console by console
          
```

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```

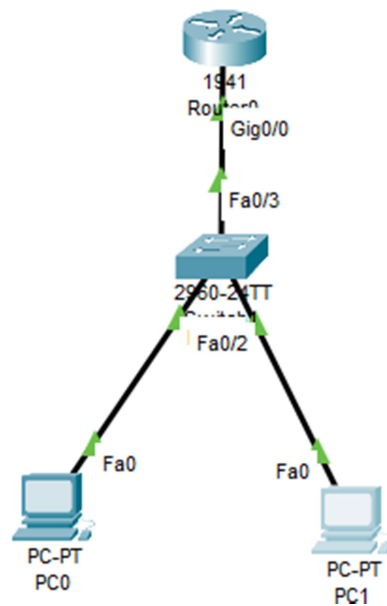
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
          
```

Fig :ping test between PC0 and PC1



```

Deep(config)#ip dhcp excluded-address 170.160.1.1
Deep(config)#ip dhcp excluded-address 170.160.1.31
Deep(config)#ip dhcp pool lan_pool
Deep(dhcp-config)#network 170.160.1.0 255.255.255.22
Deep(dhcp-config)#default-router 170.160.1.1
Deep(dhcp-config)#dns-server 8.8.8.8
Deep(dhcp-config)#domain-name pncampus.edu.np
Deep(dhcp-config)#ex
Deep(config)#ex
Deep#
%SYS-5-CONFIG_I: Configured from console by console
  
```

Deep#write

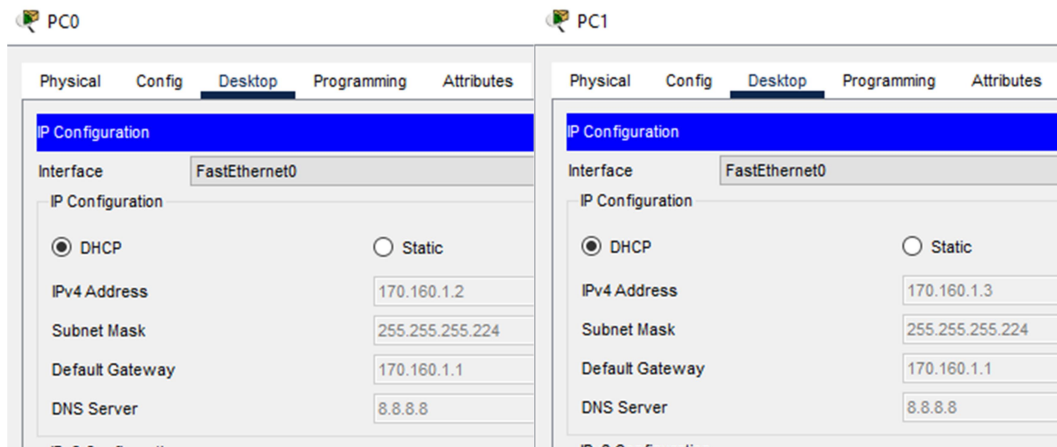


Fig: Ip addresses in PCs